

MUHAMMED MUIZ ARUMMAL

AI Engineer

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PROFILE

AI engineer who builds and evaluates LLM and multi-agent systems, from the data pipeline to a deployed product. MSc AI & Machine Learning at the University of Birmingham, based in the UK, with a security background (CompTIA CySA+). Focused on production reliability: shipping the product and the evaluation that proves it works.

EDUCATION

University of Birmingham — MSc, Artificial Intelligence & Machine Learning Sept 2026 – Sept 2027 (in progress)

Modules: Intelligent Robotics, Natural Language Processing, Evolutionary Computation.

- Research focus: reliability and security of LLM evaluation and agentic systems (preprint in preparation).

TKM Institute of Technology — B.Tech, Computer Science & Engineering

CGPA 7.56/10

Modules: Artificial Intelligence, Data Structures, Algorithm Analysis & Design, Computer Networks. ACM student chapter; undergraduate team projects.

RESEARCH EXPERIENCE

Agentic-AI attacker benchmark (finbot-bench)

github.com/secslayer/finbot-bench

- Built an evaluation harness in which local LLMs (**Ollama**) act as autonomous attackers against a live, deliberately vulnerable multi-agent application (**OWASP FinBot CTF**), scored by the target's own event-driven exploit detector rather than by self-judging.
- Compared 3 attacker architectures (single-shot, agentic loop, multi-agent); architecture, not model size, determined success.

Judge-Shaped, Not Harm-Shaped: jailbreak-judge validity

preprint in preparation | github.com/secslayer/jlens-fuzz

- Showed that the widely reused **hubert233/GPTFuzz** judge inflates reported attack-success rates by rewarding jailbreak-*shaped* role-play framing over actually harmful content.
- Designed a stricter two-stage judge (refusal pre-filter + anti-roleplay rubric); documented its residual failure (n=5 pilot).

Reproduction of a 2026 adaptive-jailbreak framework

github.com/secslayer/reproductions

- Reproduced arXiv:2505.23404 end to end on 3 local models, with an ablation and the adaptive defence the authors named but never tested; also reproduced **GPTFuzzer**.
- Found grader errors in both directions; re-verified every verdict by hand and published the corrections.

Authorization audit of a deployed multi-agent (A2A) platform

security audit & fix

- Audited against Ferrag et al.'s 14-row agent-protocol threat matrix (*ICT Express* 2025), with static + runtime evidence.
- Found and fixed a missing authorization check on the agent-registry write path; verified by regression test (401 unauthenticated / 403 wrong permission / 201 authorised).

PROJECTS

Multi-agent AI assistant platform (GptGram)

github.com/secslayer/Gptgram-production-ready

- Built and deployed a multi-agent LLM platform on **AWS** (EC2, S3): agents collaborate on each query, with **RAG** over **Qdrant** for grounded answers and conversation state persisted to S3.
- Served concurrent inference through **REST APIs** over distributed **Celery** job queues, under Git and CI/CD.

AI-driven security monitoring (Secure Guard)

github.com/secslayer/ai-soc

- Built a telemetry pipeline (Sysmon/Winlogbeat → Logstash → **Elasticsearch**, cleaned in **Python** and **SQL**) normalising **45K+** security events.
- Trained **XGBoost** and **SVM** malware classifiers with PCA-tuned cross-validation (**~85% accuracy**); used **Prophet** forecasting to turn alert thresholds into response playbooks.

Custom GPU kernels (personal project)

CUDA | cuTile

- Implemented **CUDA** kernels for core PyTorch operators to study GPU performance, then re-implemented them in **cuTile** on its release; benchmarked against PyTorch's native kernels, which remained faster.

Also built: **Ggram**, an agentic platform using **graph neural networks** to route and solve ML tasks; a **scikit-learn** SOC chatbot combining several models that cut simulated false positives by **~30%** (proof of concept, not public).

TECHNICAL SKILLS

- **Languages:** Python, C++ (incl. multithreading), SQL; CUDA (incl. cuTile) GPU programming.
- **AI & ML:** pandas, NumPy, scikit-learn, PyTorch, TensorFlow, JAX, XGBoost; generative AI & LLMs, agentic & multi-agent systems, RAG, prompt engineering, model evaluation, LLM red-teaming, prompt injection.
- **Cloud & data:** AWS (EC2, S3), Docker, Linux, Git, CI/CD, REST APIs, network programming, Databricks, Power BI; Agile, Scrum.
- **Tools & frameworks:** LangChain, Qdrant, Ollama, Celery; Grafana, Prometheus.
- **Professional:** clear technical writing for non-specialists, honest reporting of results, including negative ones.
- **Security:** detection engineering, SIEM (ELK, Splunk), incident response, threat hunting, MITRE ATT&CK.

CERTIFICATIONS

- **CompTIA CySA+ (CS0-003)** — [verify](#) | **Mathematics for Machine Learning**, Imperial College London (Coursera) — [verify](#) | **SOC Analyst**, TryHackMe — [certificate](#)

- **In progress:** AWS Certified Generative AI Developer (AIP-C01), exam not yet sat | HTB Certified Offensive AI Expert (COAE).

Kaggle: Gemma 4 Developer Agent (2026): autonomous coding agent with a failure-classifying local evaluation loop; AI Agent Security — Multi-Step Tool Attacks (OpenAI, Google & IEEE, 2026): red-teaming tool-using AI agents.